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|                   | <b>ADVANCED BUSINESS<br/>ANALYTICS USING SPSS</b> | <b>L</b> | <b>T</b> | <b>P</b> |
| <b>Credits: 4</b> |                                                   | <b>4</b> | <b>-</b> | <b>-</b> |

## **MBA – II Semester**

### **Course Objectives:**

1. To enable students on how to harness the potential of data analysis using SPSS, and turn raw data into actionable insights that drive informed decision-making.
2. To provide hands-on experience to students in data preparation, analysis, and visualization, equipping you with the expertise needed for real-world research and business applications.
3. To facilitate the knowledge of SPSS effectively, design robust studies, and troubleshoot data issues, making you a proficient data analyst.
4. To apply SPSS based skills for a wide range of statistical techniques, from correlation and regression to hypothesis testing and beyond, empowering you to tackle diverse data challenges.
5. To advance career prospects for students by adding a valuable skill set to your portfolio indispensable in data-driven industries and research environments.

### **Module 1 Getting to know IBM SPSS statistics:**

Starting SPSS statistics, opening an existing data file. Work with data files, windows, menus, dialogue boxes, and closing IBM SPSS statistics. Getting help from IBM SPSS. Introduction to samples and the population, level of measurement a special case: rating scales, independent and dependent variables, data access. Data checking by using minimum, maximum and number of valid cases, identifying inconsistent responses, when errors are discovered, SPSS missing values option

### **Module 2 Preliminary Data Analysis and Plotting:**

Exploratory Data Analysis: Interval Scale Data – Frequency Tables and Histograms, Exploratory Data Analysis, Options with Missing Values, Measures of Central Tendency, Variability Measures, Confidence Band for Mean, Shape of the Distribution, Stem & Leaf Plot, Box & Whisker Plot, Saving an Updated Copy of the Data. Mean Differences Between Groups I: (Simple Case) Logic of Testing for Mean Differences, Sample Size, Exploring the Different Groups, T – Test, Displaying Mean Differences, Paired T Test, Normal Probability Plots.

### **Module 3 Statistical Techniques to Compare Groups:**

Mean Differences Between Groups II: (One Factor ANOVA) Logic of Testing for Mean Differences, Factors, Exploring the Data, Running One-Factor ANOVA, One-Factor ANOVA Results, Post Hoc Testing of Means, Graphing the results. Mean Differences between Groups III:

(Two Factor ANOVA): Logic of Testing and Assumptions, Interactions, exploring the data, Two-Factor ANOVA, The ANOVA Table, observed means, presenting the results.

#### **Module 4 Multivariate Statistics:**

Reading the Data, Exploring the Data, Scatterplots, correlations. Introduction to Regression: The Regression Equation and Fit Measure, Residuals and Outliers, Assumptions, Simple Regression, Multiple Regression, Residual Plots, Multiple Regression Results, Residual and Outlier Results, Summary of Regression Results, Stepwise Regression, Stepwise Regression Results, Stepwise Summary.

#### **Module 5 Advanced IBM SPSS Techniques to Explore Relationships:**

Data view tabs in SPSS, Variable view tabs in SPSS, Entering data, saving a file, Select cases, Merging, Splitting file, Export and Import, Graphs, Comparing Groups (Categorical Data) A Basic Two-Way Table, Chi – Square Test of Independence, Requesting the Chi-Square Test, Different Tests and Different Results. Association Measures Available within Crosstabs, Graphing Cross Tabulation Results, Three-Way Tables creation.

#### **Suggested Readings & References**

##### **Text Books:**

1. S.P. Gupta: Statistical Methods, Sultan Chand & Sons
2. Anil Kumar Mishra - A Hand Book on SPSS for Research Work, 1st Edition, Himalaya Publishing House, 2011.

##### **Reference Books:**

1. Business Statistics: S. L Aggarwal, S. L. Bhardwaj, Kalyani Publications, 2015.
2. Fundamentals of Statistical: S. P Gupta, Sultan Chand, 2017
3. Business Statistics: J. K. Sharma, Vikas Publishers, 2014.
4. Darren George & Paul Mallery - SPSS for WINDOWS step by step, 8th Edition, Pearson Education, 2009.

##### **Course Outcomes:**

**At the end of the course, the student will be able to:**

- Understand data collection, managing, and cleaning data, ensuring its accuracy and reliability using SPSS
- Evaluate statistical methods and perform a wide range of statistical analyses, from basic descriptive statistics to advanced regression and factor analyses.
- Assess advanced data visualization techniques including charts and graphs, to effectively communicate your findings.
- Apply research skills, such as designing surveys, coding responses, and conducting hypothesis tests.
- Execute decision-making with confidence to make data-driven decisions, whether in academic research, business analytics, or other fields.